The second law

antocciphysics | TEACHER KEY | CPA

Learning target: Explain the direction of heat transfer and the limits of energy conversion.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 5, Day 5; 3 course days later.*
No. Equilibrium must lie between 13°C and 55°C under the assumptions. Check readings or an unmodeled input.

02. Explain a worked example (worked)
Net heat flows hot to cold; equilibrium has equal temperatures under suitable isolation.

03. Look and explain (model)
Hot to cold; temperatures move toward a common value if isolated.

04. Build the solution (practice)
25 J by energy conservation.

05. Use the idea (reason)
A cyclic heat engine cannot convert all heat from a single reservoir to work; some heat rejection/entropy accounting is required.

06. Choose a method (mixed) *Review: Unit 4, Day 18; 12 course days later.*
4 J remains unaccounted. Seek another storage/transfer or check measurements.

07. Exit check (exit)
No; refrigerator is not passive/spontaneous cold-to-hot transfer; work enables it.

08. STRETCH: Apply it another way (stretch)
External work powers the transfer; it is not an isolated spontaneous cold-to-hot flow.

The second law

antocciphysics | TEACHER KEY | HONORS

Learning target: Explain the direction of heat transfer and the limits of energy conversion.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 5, Day 5; 3 course days later.*
Water gains 21000 J; metal c≈1167 J/(kg·°C). Equal energy loss/gain applies under the assumptions.

02. Explain a worked example (worked)
Net heat flows hot to cold; equilibrium has equal temperatures under suitable isolation.

03. Represent and interpret (model)
Hot to cold; temperatures move toward a common value if isolated.

04. Solve and justify (practice)
250 J; COP 4. COP is not a heat-engine efficiency and may exceed 1.

05. Analyze the reasoning (reason)
A cyclic heat engine cannot convert all heat from a single reservoir to work; some heat rejection/entropy accounting is required.

06. Choose a method (mixed) *Review: Unit 4, Day 18; 12 course days later.*
Energy=7.2 J; average opposing force≈9 N.

07. Exit check (exit)
Energy conserved; temperature-difference opportunity for work is reduced/eliminated at equilibrium.

08. STRETCH: Apply it another way (stretch)
280 J rejected; includes both removed heat and input work.